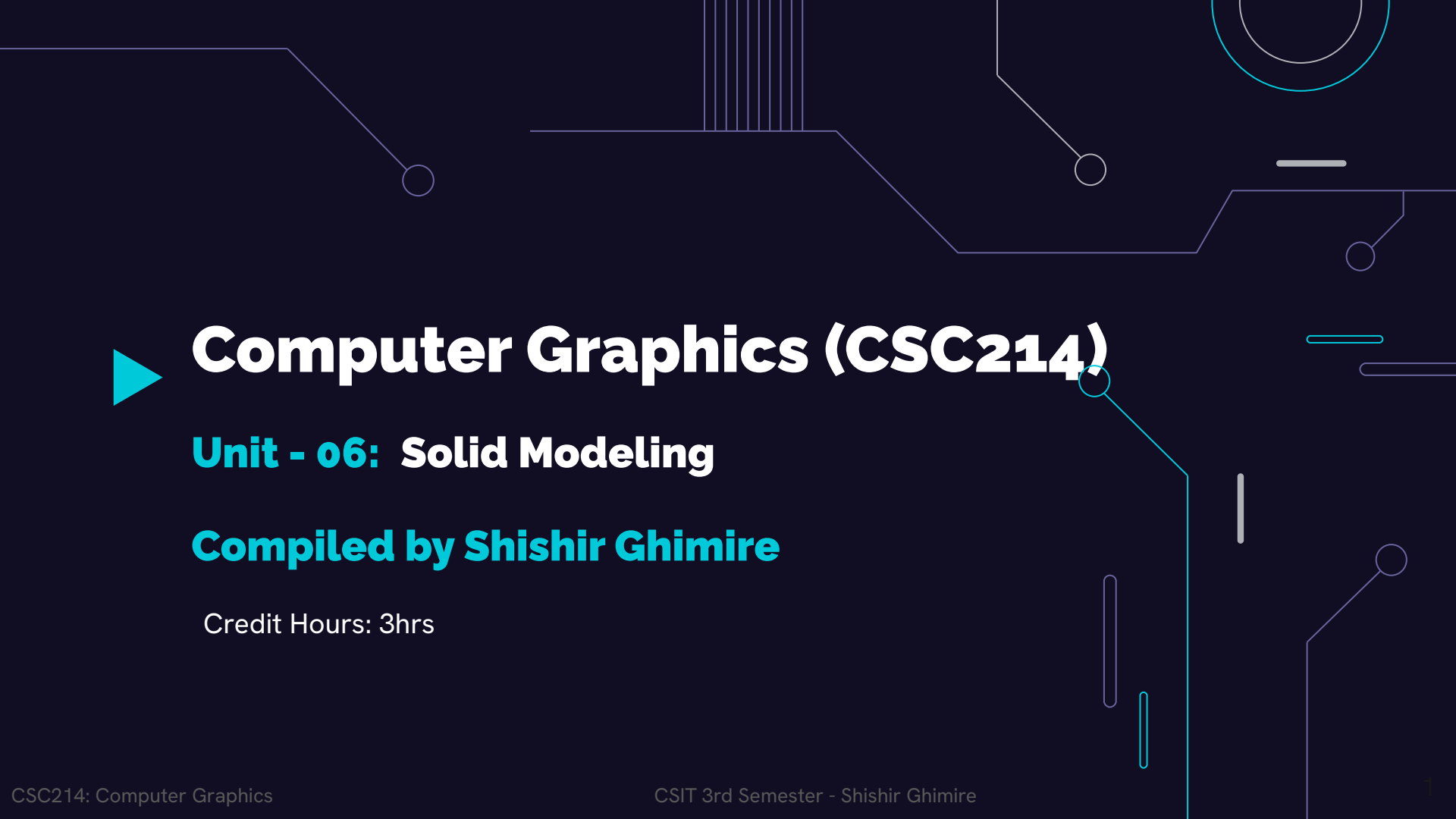




► Computer Graphics (CSC214)

Unit - 06: Solid Modeling

Compiled by Shishir Ghimire

Credit Hours: 3hrs

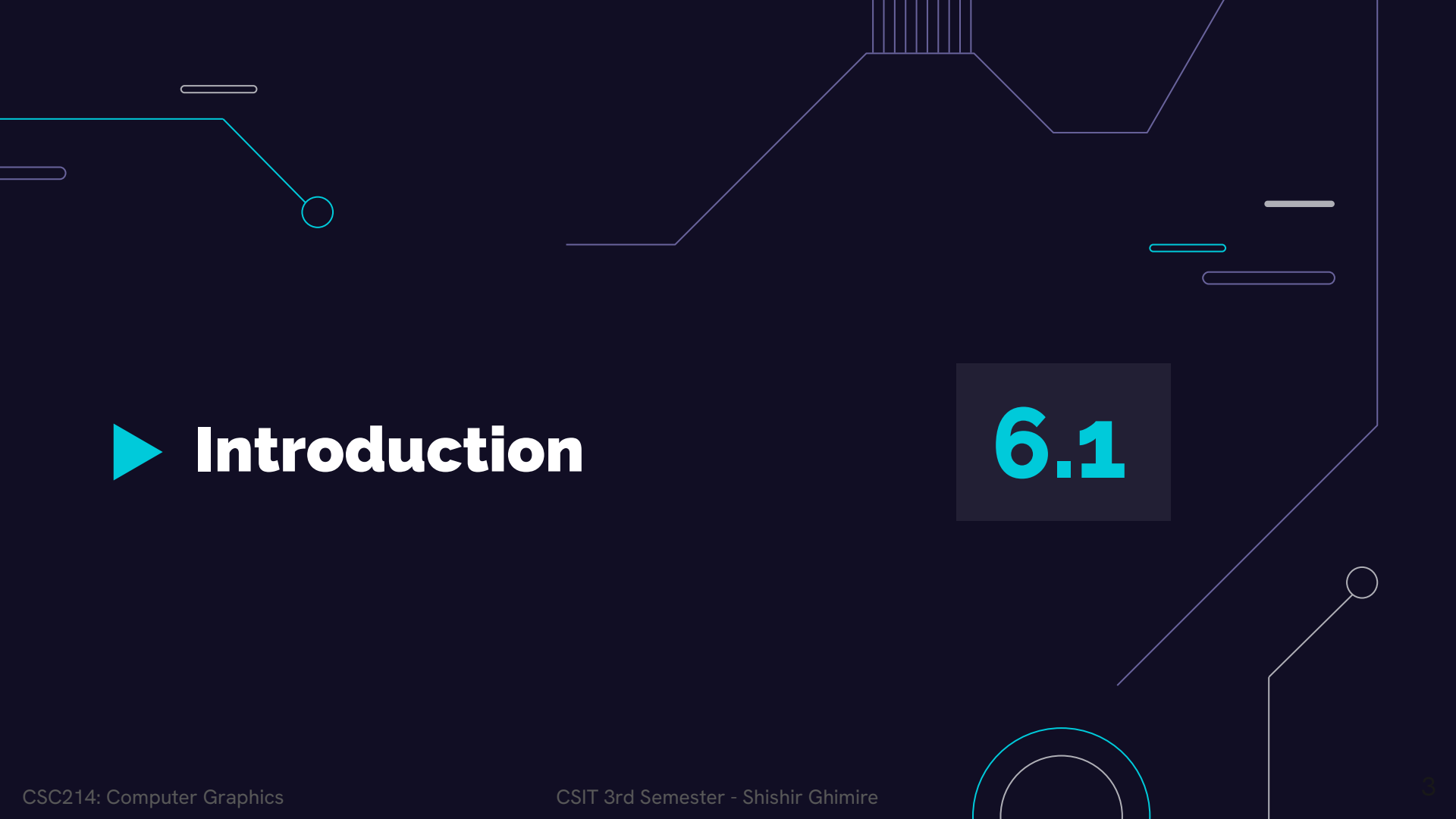
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6.2 Binary Space Partition Trees (BSP)

6.3 Octree Representation

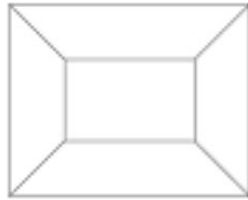


► Introduction

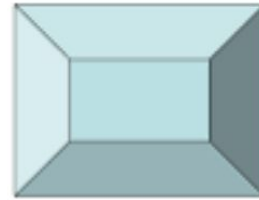
6.1

► Solid Modeling:

- ❖ There are three types of 3D model:
 - **Wireframe** - Uses lines, arcs and curves, as if the object shape were made with pieces of wire.
 - **Surface** - Uses surfaces to show the **outside of the model**, as if it were made from pieces of stretchy paper. It's **hollow inside**, but looks realistic on the outside.
 - **Solid** - Uses solid material throughout, **not hollow**.
- ❖ Solid model of an object is a **more complete representation** than its surface / wireframe model. It provides more **topological information** in addition to the **geometrical information** which helps to represent the solid **unambiguously**.



Wireframe Model



Solid Model

► Solid Modeling:

❖ Geometry vs Topology:

- **Geometry:** Metrics and dimensions of the solid object. **Location of the object** in a chosen **coordinate system**.
- **Topology:** Combinational information like **connectivity, associativity, and neighborhood** information. Invisible relationship information.

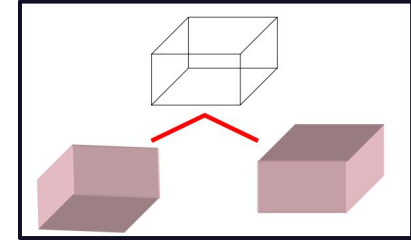
❖ Solid Modeling

- Solid modeling is a consistent set of principles for mathematical and computer modeling of three-dimensional solids.
 - A **mathematical technique** for representing solid objects.
 - An object with following specification is called solid:
 - **Vertices, edges, surfaces, weight, and volume.**

► Solid Modeling:

❖ Solid modeling is based on **complete, valid and unambiguous** geometric representation of physical object.

- **Complete:** points in space can be classified. (inside/outside).
- **Valid:** Vertices, edges, faces are connected properly.
- **Unambiguous:** there can be only one interpretation of object.



❖ Advantages of Solid Models:

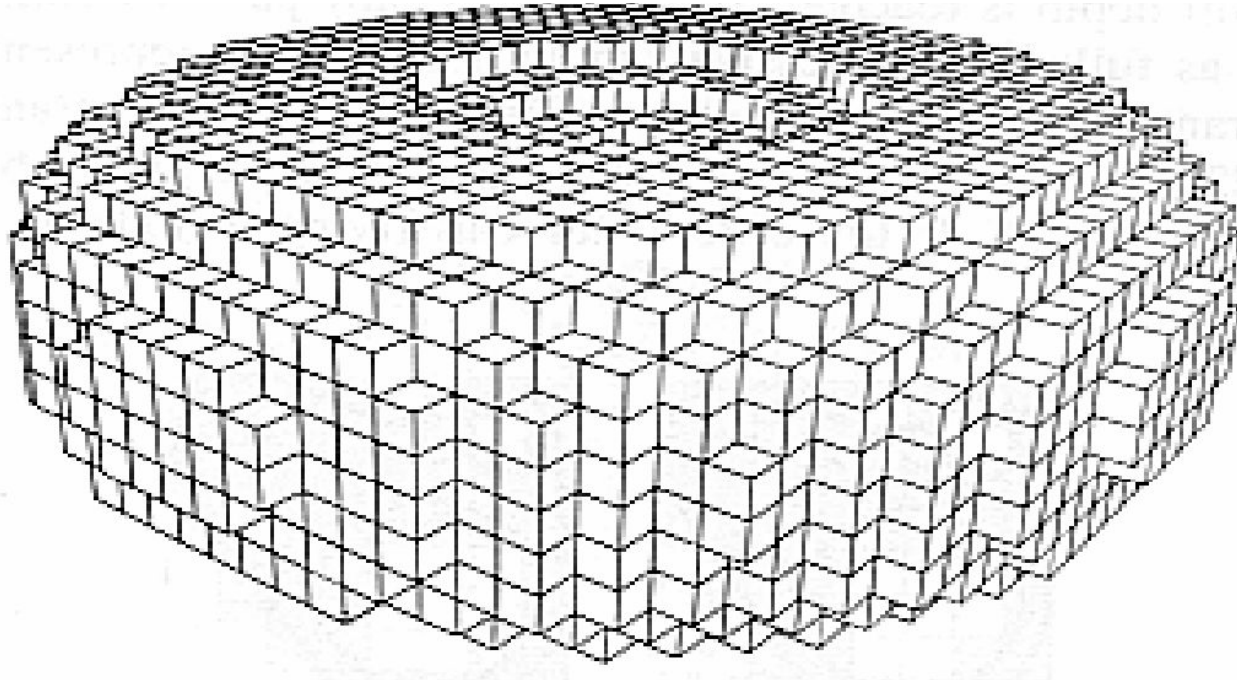
- **Unambiguous and complete representation:** Uses topological information in addition to geometrical data.
- **Accurate design:** Results in precise and reliable designs.
- **Enhances CAD/CAM goals:** Supports goals of Computer-Aided Design and Manufacturing, like Computer-Integrated Manufacturing (CIM).
- **Flexible manufacturing:** Facilitates adaptable manufacturing processes.
- **Better automation:** Leads to improved automation of the manufacturing process.

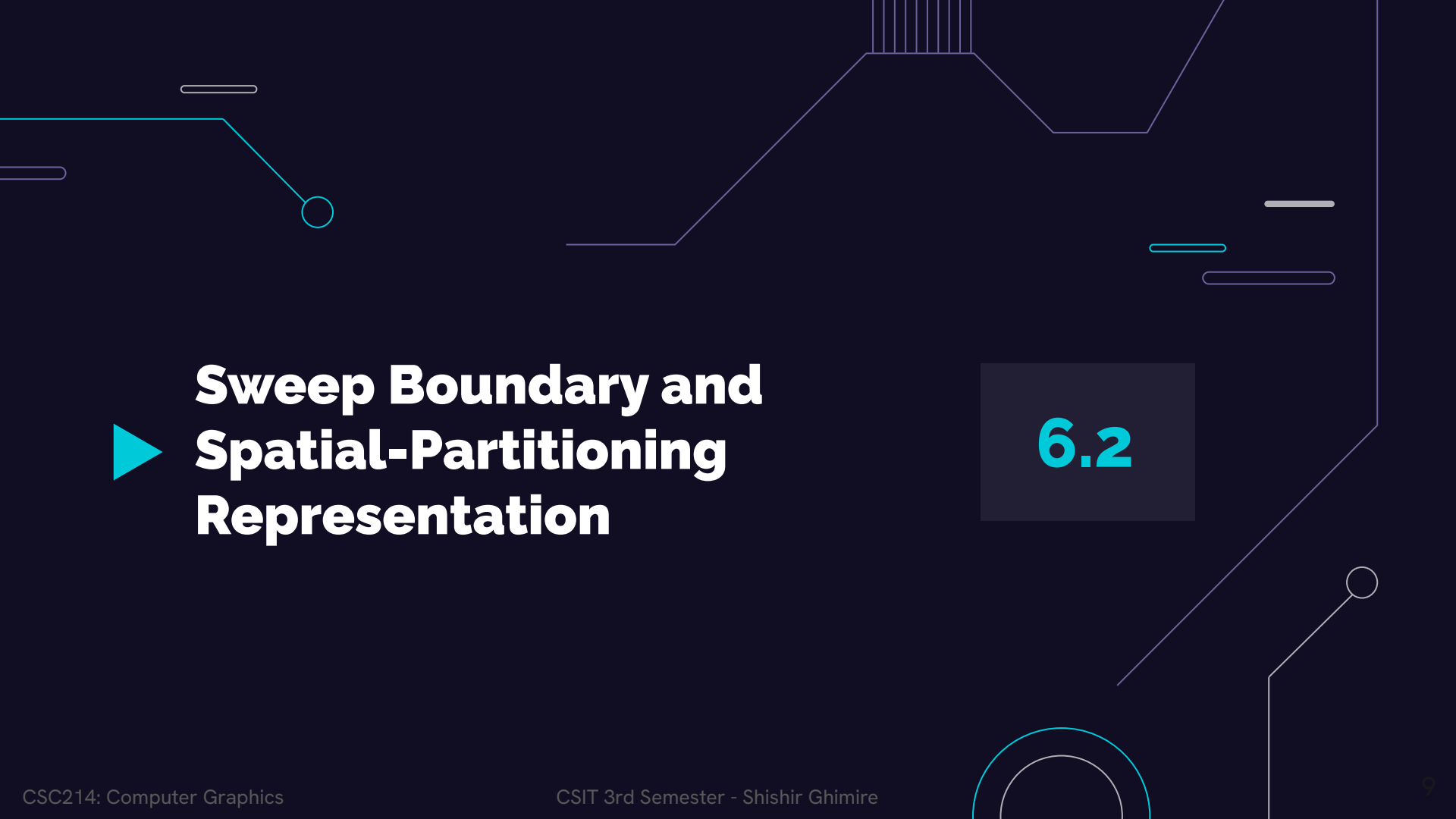
► Solid Representation Schemes:

The common representations (techniques used to create solid models) in solid modeling are:

- ❖ **Spatial Occupancy Enumeration:** In this simplest form of 3D volumetric raster model, a section of 3D space is described by a **matrix of evenly spaced cubic volume elements** called **voxels**.
- ❖ **Cell Decomposition:** This is a hierarchical adaptation of spatial enumeration. 3D space is **subdivided into cells**. Cells could be of different sizes. These simple cells are glued together to describe a solid object.
- ❖ **Boundary Representation:** The solid is **represented by its boundary** which consists of a **set of faces**, a set of **edges** and a set of **vertices** as well as their topological relations.
- ❖ **Sweep Methods:** In this technique a **planar shape** is moved along a curve. Translational sweep can be used to create **prismatic objects** and rotational sweep could be used for **axisymmetric components**.
- ❖ **Parameterized Primitive Instancing:** This modeling scheme provides a **set of possible object shapes** which are **described by a set of parameters**. Instances of object shape can be created by varying these parameters.
- ❖ **Constructive Solid Geometry (CSG):** Primitive instances are combined using Boolean set operations to create complex objects

► Solid Representation Schemes:



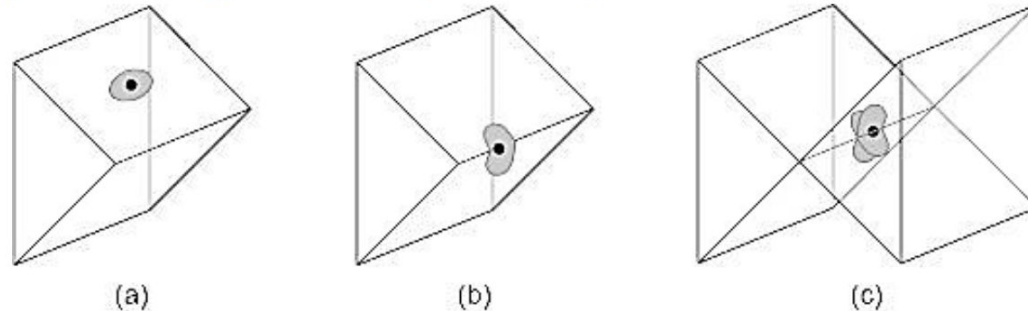


► Sweep Boundary and Spatial-Partitioning Representation

6.2

► Boundary Representation (B-rep):

- ❖ The solids are described by **its boundary surface**. Uses the description by vertices, edges and faces.
- ❖ The most common representation is the **boundary polygons**.
- ❖ Will be considered only with **solid border 2-manifolds** (wherever the point is each edge shared by two faces) i.e. the neighbors of any point of the border point are on a disk (that is to say that each edge is shared by two faces).



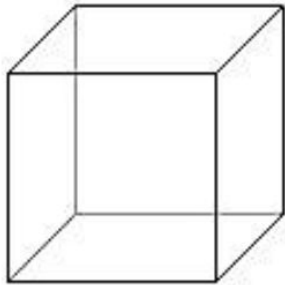
Figure(a) and (b) are **2-manifold** (c) is not **2-manifold**

► Boundary Representation (B-rep):

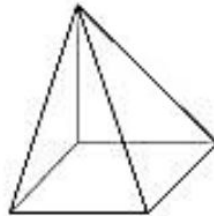
◆ Polyhedron:

- Solid delimited by a set of polygons whose **edges belonging to two polygons** (for solids 2-manifolds).

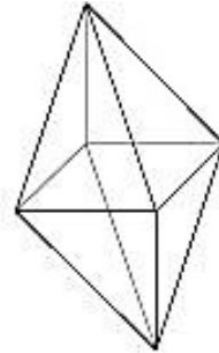
Boundary representation for single polyhedron satisfy Euler's formula
 $V-E+F=2$ where, V- vertex, E –edges & F- faces.



$$\begin{aligned} V &= 8 \\ E &= 12 \\ F &= 6 \end{aligned}$$



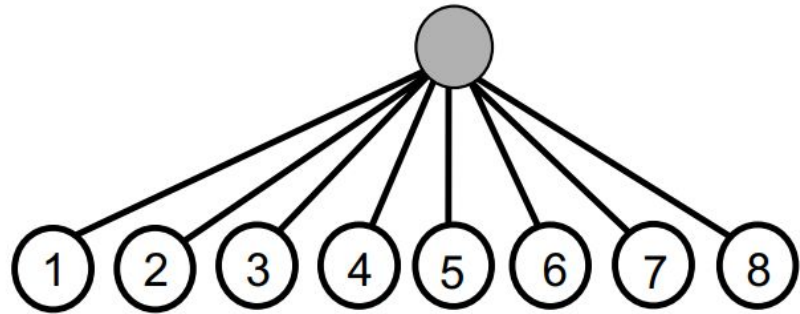
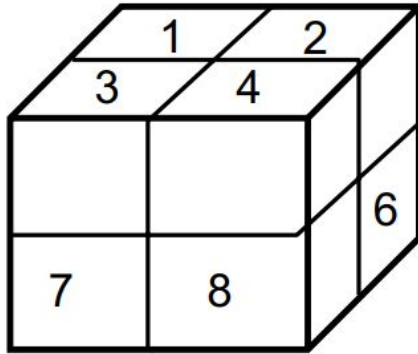
$$\begin{aligned} V &= 5 \\ E &= 8 \\ F &= 5 \end{aligned}$$



$$\begin{aligned} V &= 6 \\ E &= 12 \\ F &= 8 \end{aligned}$$

► Spatial Partitioning Representation:

- ❖ A solid is represented as a collection of adjoining **non-intersecting object**.
- ❖ For example: Octree



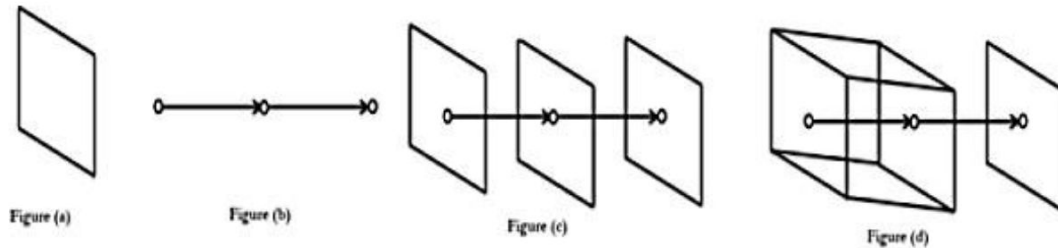
► Sweep Representation:

- ❖ Sweep representations create solid by **moving a 2D shape** (triangle, rectangle, polygon etc.) according to **predefined rule** (translating, rotating).
- ❖ Sweep representations are used to **construct 3D object from 2D shape** that have some **kind of symmetry**.
- ❖ For example, a **prism** can be generated using a **translational sweep** and **rotational sweeps** can be used to create **curved surfaces** like an ellipsoid or a torus.
- ❖ There are two types sweep representation:
 1. Traditional Sweep
 2. Rotational Sweep

► Sweep Representation:

Translational Sweep:

- ❖ A 2D shape is translated by a **predefined translational vector**.
- ❖ Example: following figure shows a translational sweep of rectangle.
- ❖ Steps:
 - Define a shape as a **polygon vertex table** as shown in figure (a).
 - Define a sweep path as a sequence of **translation vectors** figure(b).
 - **Translate** the shape; continue building a vertex table figure (c).
 - Define a **surface table** figure (d).



► Sweep Representation:

Rotational Sweep:

- ❖ A 2D shape is rotated around a **predefined rotational axis**.
- ❖ Example: following figure shows a rotational sweep.
- ❖ Steps:
 - Define a shape as a polygon vertex table as shown in figure (a).
 - Define a sweep path as a sequence of rotations.
 - Rotate the shape; continue building a vertex table as shown in figure (b).
 - Define a surface table as shown in figure (c).

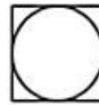
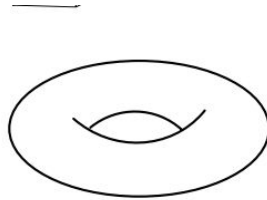
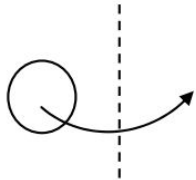


Figure (a)

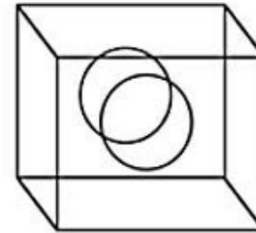


Figure (b)

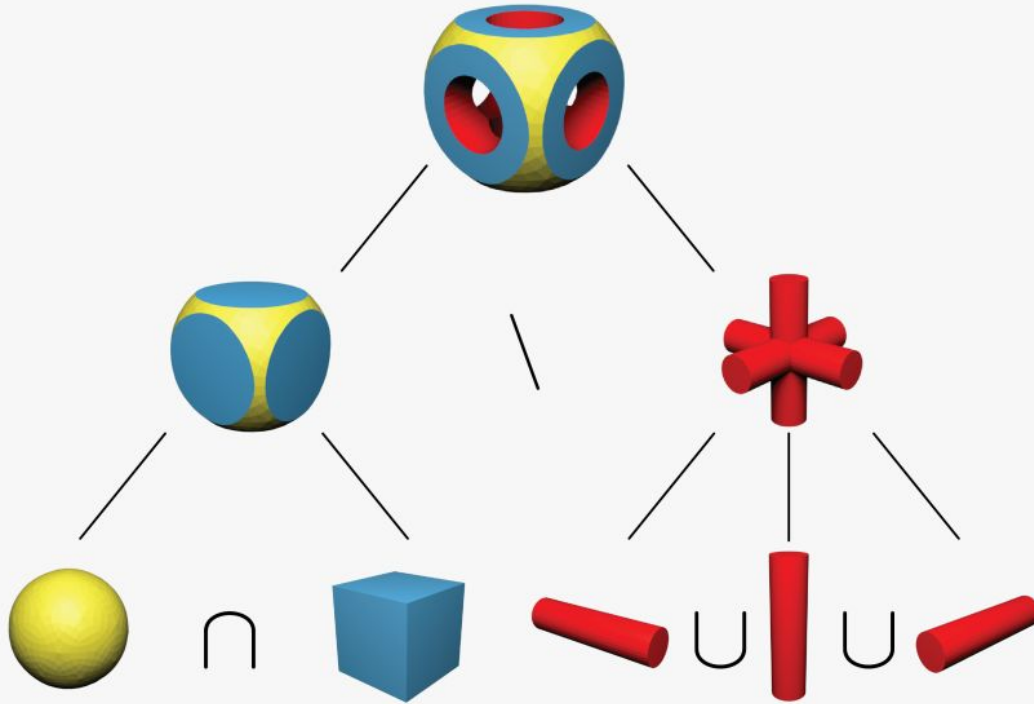


Figure (c)

► Constructive Solid Geometry (CSG):

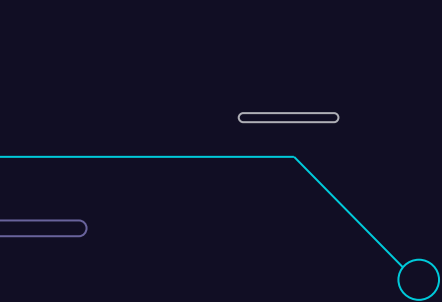
- ❖ A CSG model is based on the topological notion that a **physical object can be divided into a set of primitives** (basic elements or shapes) that can be combined in a certain order following a set of rules (Boolean operations) **to form the object**. Each primitive is bounded by a set of surfaces; usually closed and orientable.
- ❖ It is based on the idea of providing a **set of predefined object types** such as cubes, cone, sphere etc.
- ❖ A solid model is created by **retrieving primitive solids** and **performing Boolean operations**.
- ❖ Three types of Boolean operations:
 - **Union (join):** the operation combines two volumes included in the different solids into a single solid.
 - **Subtract (cut):** the operation subtracts the volume of one solid from the other solid object.
 - **Intersection:** the operation keeps only the volume common to both solids.

► Constructive Solid Geometry (CSG):

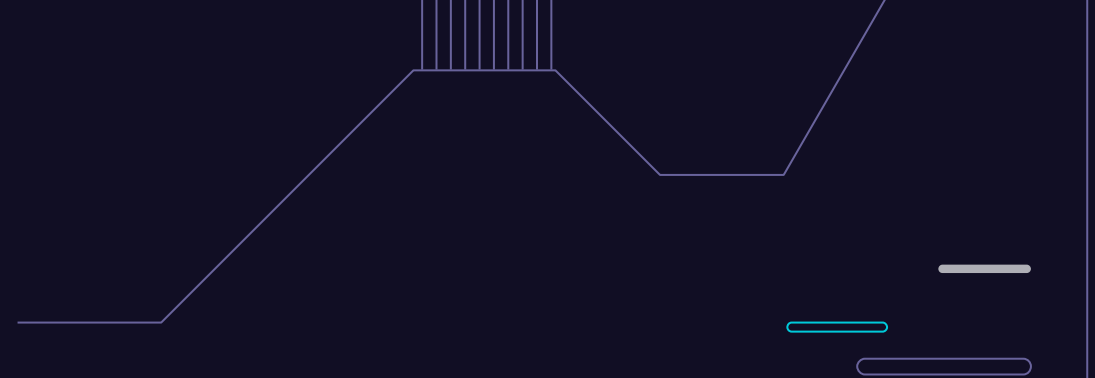


► Process of Making the Solid Models:

- ❖ **Create Wireframe Model:** Start by designing the wireframe model of the object.
- ❖ **Convert to 3D View:** Transform the wireframe model into a 3D view.
- ❖ **Add Surfaces:** Apply surfaces to the 3D wireframe model to form a 3D solid model.
- ❖ **Use CAD Software:** Specialized CAD software is required for solid modeling.
 - **Popular CAD Software:** SolidWorks (latest version: SolidWorks 2009), AutoCAD, and others.



▶ Binary Space Partitioning Tree (BSP)



6.3

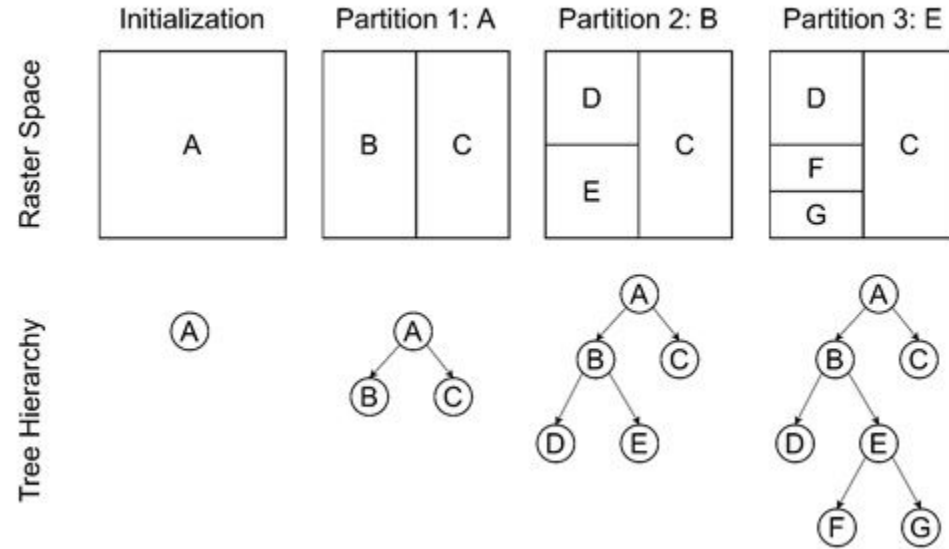
► Binary Space Partitioning Tree (BSP):

- ❖ Binary space partitioning is a generic **process of recursively dividing a scene into two** until the partitioning **satisfies one or more requirements**.
 - At each step, the space is divided by a **plane of arbitrary position** and orientation.
 - Each internal **node** of the tree is associated with a plane and has **two child pointers** (one for inside the polygon and the other to the outside).
 - If the subspace is **homogenous** (fully indoors and outdoors), cease to be divided.

► Binary Space Partitioning Tree (BSP):

- ❖ It is a type of spatial partitioning scheme where the space is divided into two halves using a plane.
- ❖ Each node in the BSP tree represents a partitioning plane, and the two child nodes represent the space on either side of the plane.
- ❖ The key idea behind BSP trees for visibility is that the scene is pre-processed into this tree structure, which can then be used to determine the visibility order of objects for any viewpoint.

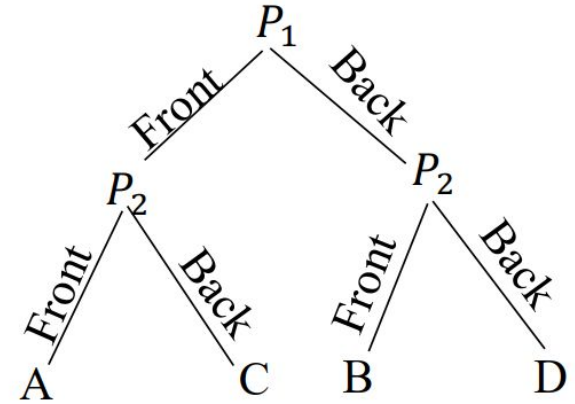
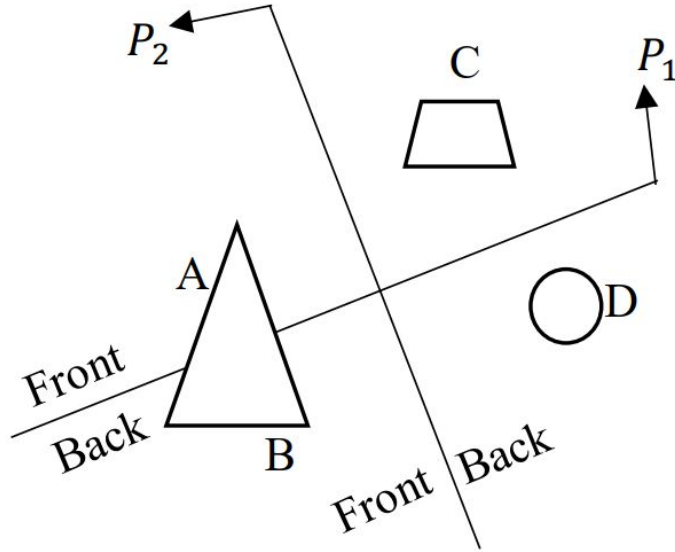
► Binary Space Partitioning Tree (BSP):



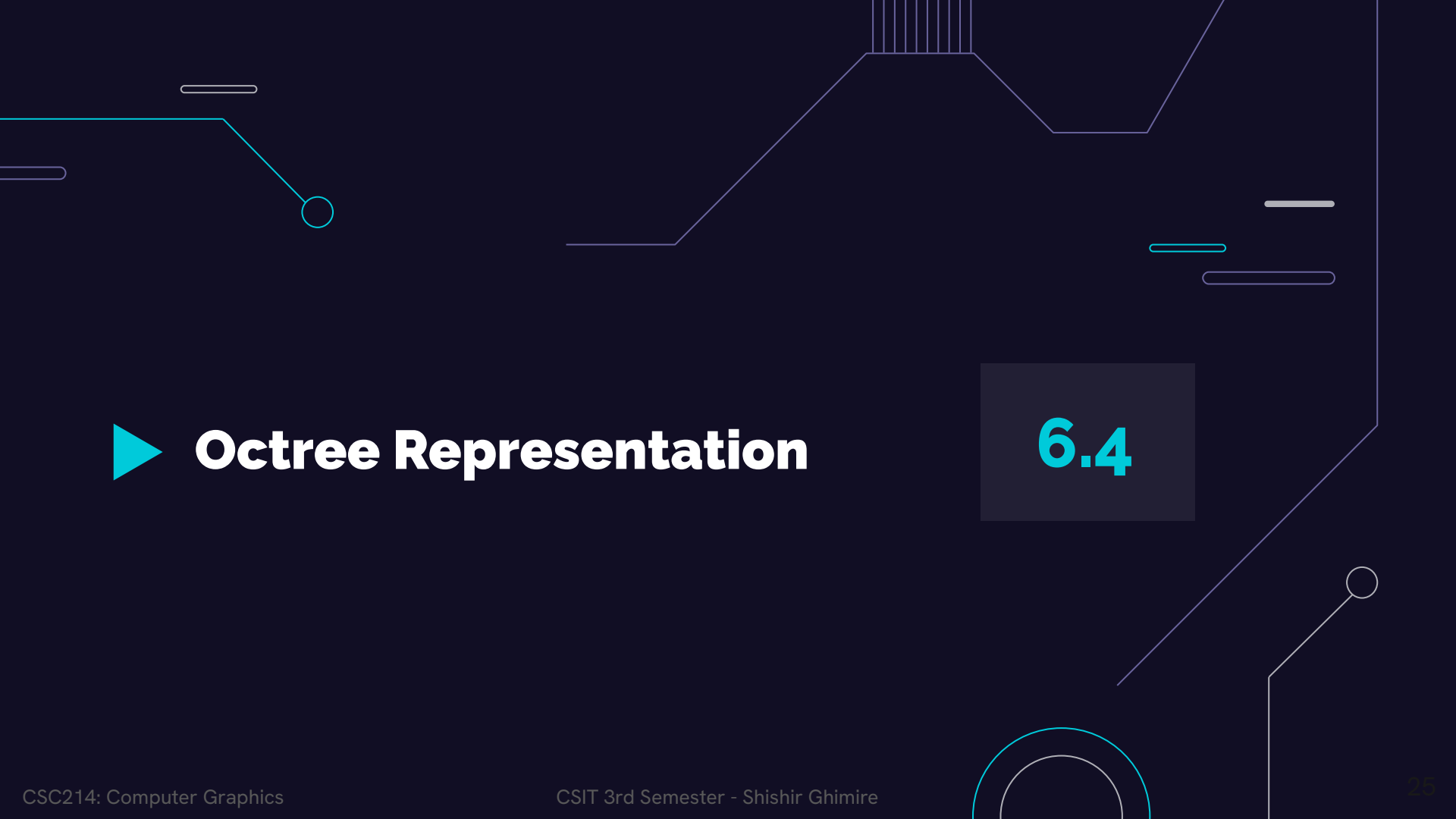
► BSP Tree Method:

- ❖ Binary Space Partitioning (BSP) tree is an efficient method for determining visibility when view reference point changes but the scene are at fixed position.
 - It involves identifying surfaces that are inside/front and outside/back w.r.t. the partitioning plane at each step of the space division, relative to the viewing direction.
 - Start with any plane and find one set of objects behind and the rest in the front.
 - In the tree, the objects are represented as terminal nodes with front objects as left branches and back objects as right branches.
- ❖ This algorithm partitions the space into sub divisions until all object list contain no more than one object.

► BSP Tree Method:



BSP tree



► Octree Representation

6.4

► Octree Representation (Solid-object representation):

- ❖ This is the **space-partitioning method** for 3D solid object representation. Octrees are hierarchical tree structures that describes each region space as nodes. They are used to represent solid objects in some graphics system.
 - **Medical imaging and other applications** that require displays of object cross sections commonly use octree representation. E.g. CT-scan.
- ❖ Octrees are used to partition a 3D space by **recursively subdividing it into eight octants**. Octant subdivisions continue until the region of space contains **only homogeneous octants**.
 - Octrees are often used in 3D graphics and 3D game engines.

► Octree Representation (Solid-object representation):

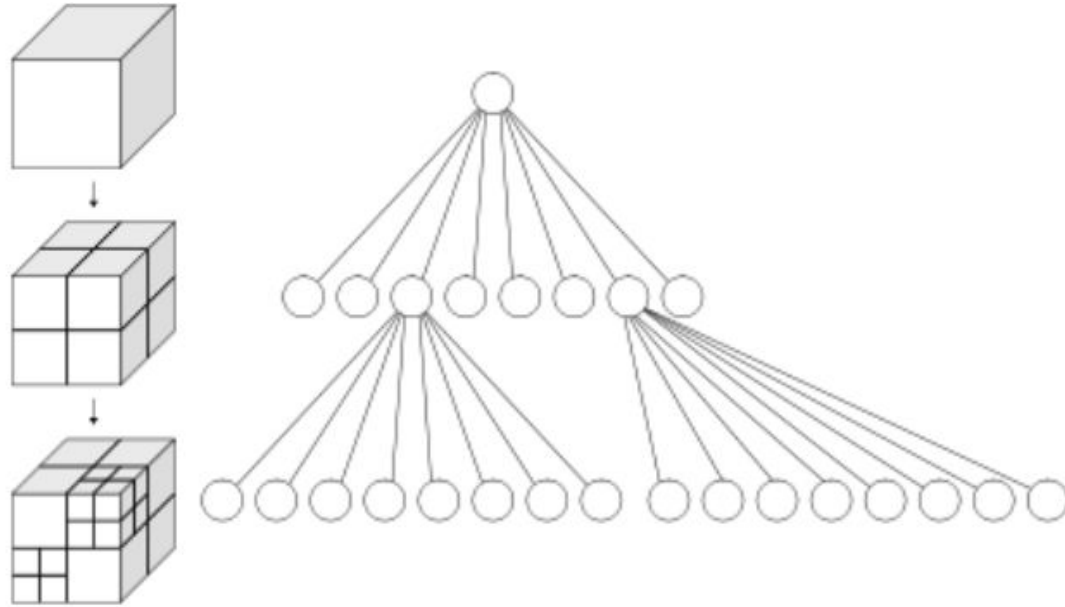


Fig: Left: Recursive subdivision of a cube into octants & Right: The corresponding octree

► Octree Representation (Solid-object representation):

❖ Advantages:

- It provides a convenient representation for storing information about object interiors.
- They can represent arbitrary shapes and that we can quickly analyze what is present at a specific position in space.

❖ Disadvantages:

- Imprecise representation, high storage demands and complex transformation operations.

► Assignment -6:

Deadline: February 17, 2026

- ❖ What do you understand by solid modeling? Explain binary space partition method. 2080
- ❖ What is the method to recognize boundary point and interior point in solid modeling? Describe how BSP recursively subdivided a space into convex sets. 2078
- ❖ Describe how a polygon can be represented using BSP tree with example. 2077
- ❖ Write short notes on (any two) 2079
 - BSP Tree
 - Octree
 - Types of Sweep
- ❖ What are the advantages and disadvantages of B-rep?
- ❖ Write short notes on CSG.
- ❖ List all the common techniques used to create solid models.



▶ **THANKS!**

Do you have any questions?

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